C 23370 **(Pages : 3)** **Name.....**

Reg. No.....

SECOND SEMESTER M.Sc. DEGREE [REGULAR/SUPPLEMENTARY] EXAMINATION, APRIL 2022

(CBCSS)

Physics

PHY 2C 08—COMPUTATIONAL PHYSICS

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

General Instructions

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section / Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section / sub part / sub division may be ignored.*
4. *There will be an overall ceiling for each Section / Part that is equivalent to the maximum weightage of the Section / Part.*

Section A

(8 Short questions answerable within 7.5 minutes)
(Answer **all** *questions, each carry weightage* 1*)*

1. Explain the numeric data types in python programming.
2. What are global variables?
3. Write a python program to plot a sine wave from 0 to 2π.
4. What are the steps involved for reading a text file in python ?
5. Explain the process of curve fitting.
6. Give the basic differences between initial value and boundary value problems.
7. Write a program to create a NumPy array of integers {1, 2, 3, 4, 5}.
8. What is a dictionary in python ?

(8 × 1 = 8 weightage)

Turn over

Section B

(4 essay questions answerable within 30 minutes)
(Answer any **two** *questions, each carry weightage 5)*

9. Explain the least square curve fitting method for a polynomial of nth degree.

10. Using Newton's forward difference formula, derive a general formula for numerical integration and hence establish Simpson's one-third rule.

11. Explain the Runge-Kutta method of fourth order. Using this method, evaluate the value of y (0.2) for the function :

$$\frac{\partial y}{\partial x} = 1 + y^2 \; ; \; x_0 = 0 \; ; \; y_0 = 0.$$

12. Write a python program to estimate the value of π using Monte Carlo simulation method.

(2 × 5 = 10 weightage)

Section C

(7 problems answerable within 15 minutes)
(Answer any **four** *questions, each carry Weightage 3)*

13. Write a Python program to find the factorial of a number provided by the user.

14. Write a Python program to analyse the Fourier series of a triangular wave function.

15. The function $y = \sin(x)$ is tabulated below :

x	...	$y = \sin(x)$
0	...	0
π/4	...	0.70711
π/2	...	1.0

Find the value of sin(π/6) using Lagrange's interpolation formula.

16. Approximate the area under the curve, $y = \frac{1}{x}$, between $x = 1$ and $x = 5$ using the trapezoidal rule with $n = 4$ sub-intervals.

17. Use Simpson's rule with $n = 4$ to approximate the integral :

$$\int_0^8 \sqrt{x}\, dx.$$

18. Write a short note on Numerov's method in numerical analysis.

19. Write a python program to obtain the trajectory of a freely falling body using Euler method.

(4 × 3 = 12 weightage)